

Date: 2026-07-09
Request ID: 100048750

Katanax		
Analyte(s)	Mn	
SOP#		
Acid Cocktail	40 mL 1:1 HCl	
sample(s)	weight (g)	volume (mL)
200128534_1	0.2054	250
200128534_2	0.2035	250
200128535_1	0.2074	250
200128535_2	0.2053	250
200128536_1	0.2064	250
200128536_2	0.2073	250

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128534	25.9934	28.0094	27.6473	17.96	0.8204

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128534_1	100/0.5	2.011	596709.45	59.67
200128534_2	100/0.5	2.057	616057.38	61.61

mn result: 606383.41 ppm

mn result: 60.64 %

mn oxide factor: 1.291

mn oxide result: 78.28 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128535	22.7511	24.7522	24.4392	15.64	0.8436

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128535_1	100/0.5	2.024	578418.78	57.84
200128535_2	100/0.5	1.978	571055.03	57.11

mn result: 574736.91 ppm

mn result: 57.47 %

mn oxide factor: 1.291

mn oxide result: 74.20 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128536	27.1506	29.1572	28.8837	13.63	0.8637

Mn ICP analysis

sample	Dilution	conc. [mg/L]	Mn_ppm [dried]	%Mn [dried]
200128536_1	100/0.5	2.074	581709.62	58.17
200128536_2	100/0.5	2.217	619118.21	61.91

mn result: 600413.92 ppm

mn result: 60.04 %

mn oxide factor: 1.291

mn oxide result: 77.51 %

mn lot: *Manufacturer: Assurance, 28-137MNY exp: 2027-01-30*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$